7.4 + 9.3 Applications

1. Malthusian growth model

$P(t)$ — population of species at time t

$$\begin{cases} \frac{dP}{dt} = rP & ; r>0 \text{ const (reproductive rate)} \\ P(0) = P_0 \end{cases}$$

$\rightarrow P(t) = P_0 e^{rt} \rightarrow \infty$ as $t \rightarrow \infty$

2. More realistic model — logistic model

$$k\frac{dp}{dt} = rp\left(1-\frac{p}{k}\right)k \quad ; r,k>0$$

$$P(0) = P_0 \geq 0$$

$$\rightarrow \int \frac{dp}{p(k-p)} = \int \frac{r}{k}\,dt$$

$$\int \frac{1}{k}\left(\frac{1}{p} - \frac{1}{k-p}\right)dp = \int \frac{r}{k}\,dt$$

$$\frac{1}{k}\ln p + \frac{1}{k}\ln|k-p| = \frac{rt}{k} + C_1$$

$$\ln p + \ln|p-k| = rt + C$$

$$\ln\left|\frac{p}{p-k}\right| = rt + C$$

$$e^{\ln\left|\frac{p}{p-k}\right|} = e^{rt}e^{c}$$

$$\frac{p}{p-k} = \pm Ce^{rt}$$

$$p = pCe^{rt} - kCe^{rt}$$

$$kCe^{rt} = p(Ce^{rt}-1)$$

$$p = \frac{kCe^{rt}}{Ce^{rt}-1}$$

Plugging $t=0$ in $\rightarrow \frac{P_0}{P_0-k} = C$

$$\therefore \boxed{P(t) = \frac{k\left(\frac{P_0}{P_0-k}\right)e^{rt}}{\left(\frac{P_0}{P_0-k}\right)e^{rt}-1} = \frac{kP_0e^{rt}}{P_0e^{rt}+k-P_0} = \frac{P_0}{\frac{P_0}{k}+\frac{k-P_0}{k}e^{-rt}}}$$

Lost equilibrium sols: $P\left(1-\frac{p}{k}\right)=0 \rightarrow p = 0, k \rightarrow$ luckily, we don't lose them in

As $t \rightarrow \infty$ $P(t) = k$ as long as $P_0 > 0$

Rk1: If $p_0 < 0$, then $P(t)$ blows up at t s.t. $p_0 + (k-p_0)e^{-rt} = 0$

$$e^{-rt} = \frac{p_0}{k-p_0}$$

$$-rt = \ln\left(\frac{p_0}{k-p_0}\right) \rightarrow t_{blowing} = \frac{1}{r}\ln\left(\frac{k-p_0}{p_0}\right)$$

Rk2: Monotonicity of $P(t)$ if $p_0 > 0 \rightarrow e^{-rt} \searrow$ as $t \nearrow$

$$\rightarrow \frac{P_0}{k} + \frac{k-P_0}{k}e^{-rt} \quad \begin{cases} \searrow & \text{if } P_0<k \\ \nearrow & \text{if } P_0>k \\ =\frac{P_0}{k} & \text{if } P_0=k \end{cases} \quad \rightarrow P(t) = \begin{cases} \nearrow & \text{if } P_0<k \\ \searrow & \text{if } P_0>k \\ k & \text{if } P_0=k \end{cases}$$

Moral of the story: as long as $p_0 \geq 0$, $p(t) \rightarrow \boxed{k}$ increasingly/decreasingly depending on if $P_0<k$ or $P_0>k$

↳ carrying capacity of habitat

3. Newton's cooling law

$H(t)$: temperature inside at time t, H_s = temp of surrounding medium

$\frac{dH}{dt} \propto H - H_s \rightarrow \frac{dH}{dt} = -k(H - H_s)$; k = proportionality const

To solve it, let let $y = H - H_s \rightarrow \frac{dy}{dt} = -ky$

$$e^{kt}\left[\frac{dy}{dt}\right] + kye^{kt} = 0$$

$$(e^{kt}y)' = 0 \rightarrow e^{kt}y = C$$

$$y = Ce^{-kt}$$

$$H = H_s + Ce^{-kt}$$

$\rightarrow H_s$ when $t \rightarrow$ $= H_s + (H_0 - H_s)e^{-kt}$

7.4/43 Know: $H(0) = 46°C$, $H(10) = 39°C$, $H(20) = 33°C$ Want: H_s

$\therefore H(t) = H_s + Ce^{-kt}$

$46 = H_s + C \rightarrow C = 46 - H_s$

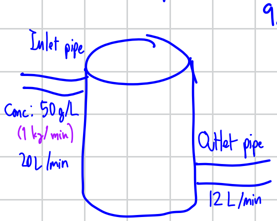
$39 = H_s + C\boxed{e^{-10k}} \rightarrow$ ☺ $\rightarrow e^{-10k} = \frac{39 - H_s}{C} = \frac{39 - H_s}{46 - H_s}$

$33 = H_s + C\boxed{e^{-20k}} \rightarrow$ ☺^{2}

$= H_s + \frac{(39 - H_s)^2}{46 - H_s}$

$33(46 - H_s) = H(46 - H_s) + (39 - H_s)^2 \longrightarrow$ แก้หา H_s ออกมา

4. Mixture problem



9.4/14 At $t = 0$, Vol of solution = 400 L

Vol of tank = 800 L

Want t_{full}, $m(t_{full})$; m_t = amount of concentrate in kg at time t

By conservation of mass, net rate of change of $m(t)$ = rate in − rate out

$$\frac{dm}{dt} = 20\ L/min \cdot 50\ g/L - \text{Conc} \cdot \text{Vol rate}$$

$$\frac{m(t)}{v(t)} \cdot 12\ L/min$$

$$\frac{dm}{dt} = 1\ kg/min - \frac{m(t)\ kg}{400 + 8t\ L} \cdot 12\ L/min$$

$\therefore 400 + 8t = 800 \rightarrow t_{full} = 50$ min

$$\frac{dm}{dt} + \frac{12}{400 + 8t}m = 1 \quad — (*)$$

$\therefore V = e^{\int \frac{12}{400 + 8t}dt} = e^{1.5 \ln|400 + 8t|} = |400 + 8t|^{3/2}$

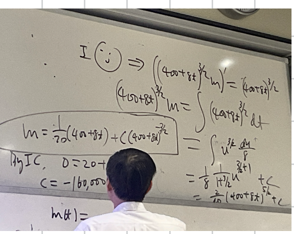
$V \cdot (*)$ $\quad ((400 + 8t)^{3/2}m)' = (400 + 8t)^{3/2}$ $\quad u = 400 + 8t \rightarrow du = 8\,dt$

$\therefore (400 + 8t)^{3/2} \cdot m = \int (400 + 8t)^{3/2}dt = \int u^{3/2}\frac{du}{8} = \frac{1}{20}u^{5/2} + C = \frac{1}{20}(400 + 8t)^{5/2} + C$

$$m = \frac{1}{20}(400 + 8t) + C(400 + 8t)^{-3/2}$$

Initial condition $m(0) \rightarrow 0 = 20 + \frac{C}{8000} \rightarrow C = -160000$

$$m(50) = \frac{1}{20}(400 + 8t) - 160000(400 + 8t)^{-3/2}\Big|_{t=50}$$



7.4/29 $\frac{dp}{dt} = rp$, $p(0) = 1$, $p(0.5) = 2p(0) \longrightarrow e^{r \cdot 0.5} = 2$

$\rightarrow p(t) = e^{rt} = e^{2\ln 2 \cdot t} = 2^{2t} = 4^t$ $\quad r = \frac{\ln 2}{0.5} = 2\ln 2$

$P(24) = 4^{24}$

9.4 Autonomous equation & Phase line analysis

$$\frac{dy}{dx} = f(y)$$ doesn't depend on x explicitly

It is separable eq, can be solved using separation of variables method, but it is time consuming.

Instead, god gives us qualitative information about solutions

Idea: visualize a solution $y(x)$ as an equation moving on the y-axis (think of x as time)

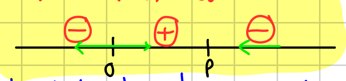
$y(x)$ — y → velocity of particle $= f(y)$

phase = state/condition of a matter in a system

Phase line analysis

e.g. $\frac{dP}{dt} = r_0(1 - \frac{P}{k})P$

Sign of $r_0(1 - \frac{P}{k})P = \frac{dP}{dt}$



Step 1: Find equilibrium sol's: $r_0(1 - \frac{P}{k})P = 0 \rightarrow P = 0, k$

Step 2: Mark equilibrium sol's on phase line. These equations sol's divide phase line into several intervals, on each of which $\frac{dP}{dt} = r_0(1 - \frac{P}{k})P$ does change sign

Step 3: Draw arrows on each interval to indicate direction of motion

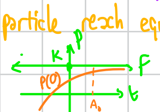
Q1: If $0 < P(0) < K$, can $P(t)$ stops before reaching k ?

Ans: No, because particle has a positive velocity

Q2: Can the particle reach equilibrium pt k in finite time? (อ้างอิง Differential equation: Uniqueness theorem)

↳ Consider $\frac{dy}{dx}$ = nice $f(x,y)$ → only 1 solution curve passing through (a,b)

Ans: No



Only 1 pt passing through A_0, K

Never stop ⇒ converse to K before K

ถ้าหยุด = มี curve สีส้มหลายอันที่เป็นไปได้

If $P(0) > k$, $P(t) \rightarrow k$ as $t \rightarrow \infty$

→ as long as $P(0) > 0$, $\lim_{t \to \infty} P(t) = k$

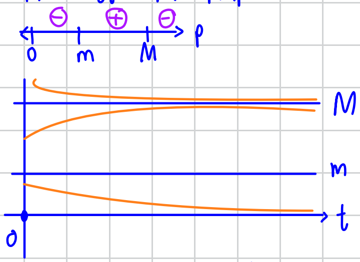
Comparison of Malthusian Model & Logistics Model

- In Malthusian model, $P(t)$ grows exponentially to ∞ as $t \to \infty$ while in logistics modal $g(t) \to k$ so long as $P(0) > 0$ as $t \to \infty$

unreasonable ↳ reasonable

Critique on Logistics Model: If $P(0)$ is too tiny, the species is supposed to extinct, yet the model predicts long time population being $\approx k$

9.4/14 $\frac{dp}{dt} = rp(M-p)(p-m)$

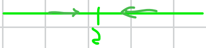


Let X = current deer population
Y = #permits (allow to kill 1 deer)

} $X - Y > m \rightarrow \underbrace{X-m}_{\text{max no. of permits issued}} > Y$

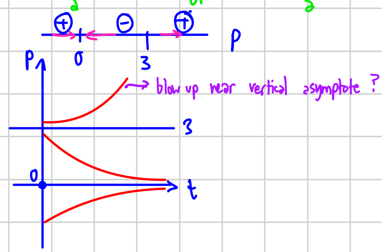
Stability of equilibrium sol's

Let a be equilibrium sol of $\frac{dy}{dx} = f(y)$

Def: If  near a, then we say a is stable

If  or  or , we say a is unstable

9.4/11 $\frac{dp}{dt} = 2P(P-3)$ (Not logistics)



Logistics model : Large P, $\frac{dP}{dt} < 0$

$P=0$: Stable

$P=3$: Unstable

HW 7.4 (30, 34, 42)
9.3
9.4